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Pedro Serrador & Adel A. Zadeh

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When Agile Works: Unveiling the Secrets to Agile Success

Pedro Serrador PhD and Adel A. Zadeh PhD

Northeastern University

ABSTRACT
While Agile methodologies are extensively advocated across various literatures as superior project management methods, empirical evidence substantiating these claims remains sparse. Furthermore, there is limited understanding regarding the optimal conditions and practices that enhance the success of Agile implementation. In our research, we analyzed a data sample of 1002 projects from various industries and countries to test the effect of Agile use in organizations. The results indicate that Agile methods are more successful than traditional methods, particularly in projects with experienced teams, medium to low stakeholder engagement, smaller team sizes, and unclear goals. However, the effectiveness of Agile methods is limited in low complexity projects with clear goals, high stakeholder engagement, large teams, or inexperienced teams. Nonetheless, teams benefit from adopting Agile methods, even if their project does not fit the ideal Agile use case. The popularity and growth of Agile methods are therefore justified.

KEYWORDS
Agile methodologies; project success; project complexity; agility

EMJ FOCUS AREAS
Continuous Improvement, New Product Development, Technology Management, Program & Project Management

Introduction

In 2001, the Agile Manifesto was introduced by a consortium of 17 software process methodologists who convened in a summit aimed at advocating for enhanced methodologies in software development, subsequently establishing the Agile Alliance (Fitsilis et al., 2014; Vierlboeck et al., 2019). The Agile movement was created to promote a flexible approach to project delivery. Since its inception, the Agile Manifesto has significantly transformed the management paradigms of software development projects, experiencing substantial adoption both vertically within organizations and horizontally across diverse business sectors (Rahy & Bass, 2022). Emergent in response to the limitations of the waterfall framework, the Agile Manifesto sought to establish a novel foundation for frameworks capable of effectively navigating escalating uncertainties inherent in contemporary development contexts (Sandstø & Reme-Ness, 2021). The 'agile manifesto' focuses on four core values (Beck et al., 2001): prioritizing individuals and interactions over processes and tools, working software over comprehensive documentation, customer collaboration over contract negotiation, and responding to change over following a plan.

Recent studies comparing traditional and Agile project management approaches have gained significant traction, reflecting the ongoing evolution in project management methodologies. The study done by Dybå and Dingsøyr (2008), compared traditional and Agile development in a number of factors, as shown in Exhibit 1. Traditional development emphasizes extensive planning, formal communication, and strict control, while Agile development focuses on rapid feedback, collaboration, and flexibility. The two approaches differ in their management style, knowledge management, development models, organizational forms, and quality control processes.

Although Agile methodologies have gained significant popularity, empirical evidence supporting their superiority over conventional project management approaches on a large scale remains limited (Calafat et al., 2022; Chandran & Das Aundhe, 2022; Kennedyd et al., 2024). Additionally, little research has been conducted on the circumstances and conditions under which Agile methods are most effective or ineffective. Therefore, this study seeks to address this research gap by analyzing a substantial dataset of international projects to examine the relationship between the use of Agile methods and project success.

In order to assess the relationship between Agile methods and project success, it is essential to first establish how success is measured. Historically, project success has been evaluated based on meeting the triple constraint of timelines, budget, and scope (Kerzner, 2017). However, the definition of project success has expanded in recent years (Alvarenga et al., 2019; Kennedyd et al., 2024; Serrador & Zadeh, 2024). Munns and Bjeirmi (1996) noted that the majority of literature considers projects to end upon delivery to the customer, with little consideration given to the broader criteria that impact the project after its deployment. While this perspective is understandable from the viewpoint of the project manager, it fails to account for the wider impact of projects on the organization, which has been the subject of research in the literature. According to Shenhar et al. (1997), scope of a project can have the greatest impact on both efficiency measures and the satisfaction of the customer, making it a crucial factor in determining project success. Jugdev and Müller (2005), reiterated this viewpoint, and noted that recent research on project success increasingly measures the

CONTACT Adel A. Zadeh a.zadeh@northeastern.edu College of Professional Studies, Northeastern University, 360 Huntington Ave, Boston, MA 02115, Canada
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Exhibit 1. Main Differences Between Traditional Development and Agile Development After Dybå and Dingsøyr (2008)

Criteria	Traditional development	Agile development
Fundamental assumption	Systems are fully specifiable, predictable, and are built through meticulous and extensive planning	High-quality adaptive software is developed by small teams using the principles of continuous design improvement and testing based on rapid feedback and Change
Management style	Command and control	Leadership and collaboration
Knowledge management	Explicit	Tacit
Communication	Formal	Informal
Development model	Life-cycle model	The evolutionary-delivery model
Desired organizational form/structure	Mechanistic (bureaucratic with high formalization), aimed at large organizations	Organic (flexible and participative encouraging cooperative social action), aimed at small and medium sized Organizations
Quality control	Heavy planning and strict control. Late, heavy testing	Continuous control of requirements, design and solutions. Continuous testing

impact of the project on the organization as a whole, rather than simply focusing on the triple constraint. Thomas et al. (2008) state that there are cases "where the original objectives of the project are not met, but the client is highly satisfied." There are other examples where the initial project objectives were met, but the client was quite unhappy with the results.

In their study, Müller and Turner (2007) defined ten dimensions of project success, with time, budget, and scope goals being the most frequently mentioned measures by participants. This suggests that aspects of efficiency remain important for organizations. Kloppenborg et al. (2009) echoed this point, noting that all views on project success should consider time, cost, and performance measures. These observations are corroborated by recent studies (Al Jabri et al., 2023; Kudyba & Cruz, 2023; Siddique & Hussein, 2022; Ullah et al., 2022) and as a result, this study will use the following constructs to evaluate project success:

- **Project efficiency** – meeting cost, time, and scope goals
- **Project success** – meeting wider business and enterprise goals

The rest of this paper provides a solid theoretical foundation that informs the research hypothesis. Additionally, it presents detailed data collection and analysis of 1,002 projects across diverse industries and countries to investigate the impact of Agile adoption in organizations. The paper then proceeds to outline the key findings from the empirical investigation of the stated hypotheses and offers recommendations to enhance the success of Agile implementation efforts

Theoretical Background and Hypothesis Formulation

This section provides a detailed explanation of the hypotheses developed for this study, which are grounded in the expectation of agile methodology's superiority, considering factors including the overall effectiveness of frameworks, project complexity, iterative planning, well-defined goals, team characteristics, and stakeholder engagement.

Overall Agile Effectiveness Project Complexity

Agile methodologies have increasingly become prevalent in project management since their inception, particularly due to their ability to enhance flexibility and responsiveness to change in complex project environments. Agile methods prioritize minimal documentation, which allows for reduced planning time and quicker implementation compared to traditional project management approaches. Recent studies, such as those by Umer et al. (2021), highlight that Agile management practices, when combined with traditional methods, can effectively mitigate the complexities faced by organizations, leading to improved project performance in the IT sector. Similarly, Koch and Schermuly (2021) emphasize that Agile practices, such as daily stand-up meetings and retrospectives, play a crucial role in managing project demands and enhancing team collaboration, which is vital for navigating the challenges posed by complex projects.

However, the effectiveness of Agile methodologies is not without its challenges. Chow and Cao (2008) conducted research on critical success factors (CSFs) in Agile project management, identifying twelve previously recognized CSFs. Their study confirmed six of these factors as essential for success: team environment (in terms of quality), team capability (in terms of timeliness and cost), customer involvement (in terms of scope), project management process (in terms of quality), Agile software engineering techniques (in terms of quality and scope), and delivery strategy (in terms of scope, timeliness, and cost). However, a subsequent study by Stankovic et al. (2013) aimed to validate Chow and Cao's findings and revealed that only half of the CSFs were confirmed by their data, indicating a gap in empirical research regarding the relationship between Agile methodologies and project success

Research by Nuottila et al. (2022) underscores the organizational hurdles that can impede the successful adoption of Agile practices, calling for further investigation into the effectiveness of Agile methods in various contexts. Furthermore, the work of Barlow et al. (2011) suggests that selecting the appropriate methodology based on project complexity is critical, advocating for Agile approaches in projects characterized by high complexity and uncertainty. This aligns with the findings of Serrador and Pinto (2015), who established a clear link between Agile methodologies and project success, reinforcing the hypothesis that Agile projects are generally more successful than traditional projects. Therefore, the following hypothesis will be examined:

Hypothesis 1: Agile projects are more successful than traditional projects.

Each project is unique, and traditional project management literature has often oversimplified this by assuming that "a project is a project." However, as noted by Shenhar et al. (2001), each project possesses distinct characteristics that must be considered to develop a successful approach. Project complexity can be assessed through various criteria, and changes such as technology upgrades or the implementation of new features may necessitate additional budget allocations for successful delivery, as highlighted by Ullah et al. (2022). The selection of an appropriate methodology is critical when initiating a project, and recent research by Demirel et al. (2017) emphasizes that choosing the right methodology is essential for project success. Their findings indicate that hybrid methodologies can be just as effective as Agile methodologies in many cases, allowing organizations to tailor their approaches to specific contexts and requirements. Moreover, the complexity of projects significantly influences the effectiveness of Agile methodologies. Tominc et al. (2023) and Shenhar et al. (2001) both observed that complexity and unrealistic planning are major contributors to software project delays and failures, indicating that complexity is a critical factor in the success of Agile projects. Barlow et al. (2011) proposed a methodology selection framework that addresses the specific needs of organizations, recommending plan-driven methodologies for projects with sequential interdependencies, Agile methodologies for small teams with reciprocal interdependencies, and hybrid methodologies for larger teams with complex interdependencies (Brühl, 2022; Mishra & Alzoubi, 2023; Mishra et al., 2021). This framework underscores the importance of coordination support, such as collocation and knowledge transfer systems, particularly for high-volatility project teams. Recent studies further support the notion that complex and volatile projects necessitate the use of Agile methodologies to enhance project success (Ciric Lalic et al., 2022; Koch & Schermuly, 2021). Consequently, the following hypothesis will be examined:

Hypothesis 2: Agile environments are more successful with projects of high complexity.

Planning

Planning in Agile projects is a dynamic and integral part of the execution phase, often involving significant adjustments and ongoing assessments (Mubarak et al., 2019). Koskela and Abrahamsson (2004) analyzed the customer's role in an XP project (an Agile software development methodology). They found that customers were highly involved in the planning game sessions, acceptance testing, and retrospective sessions at the end of release cycles. They found that planning took up 42.8% of customer's effort, followed by acceptance testing (29.9%), postmortem (13.4%), project meetings (9.2%), and coaching (5.7%), see Exhibit 2.

Al Jabri et al. (2023) emphasize that Agile methodologies incorporate planning throughout execution, necessitating a reevaluation of tasks typically reserved for the planning phase. This integration is crucial, as all plans must account for potential changes during execution, a principle supported by Gerona and Ocampo (2023), who discuss the need for a hybrid approach that incorporates both Agile and traditional planning methods to effectively manage project complexity. Agile practices, such as daily meetings, serve as structured planning sessions that facilitate continuous alignment and adaptation, as noted by Al Jabri et al. (2023), who highlights the increased attention to Agile project management since 2021, largely due to the COVID-19 pandemic. This suggests that Agile methodologies require constant planning to ensure optimal delivery, as emphasized by Elkhatib et al. (2022).

However, the relationship between planning effort and project success is complex. Serrador (2013) found a nonlinear relationship, indicating that both excessive and insufficient planning can negatively impact project outcomes (Astridita et al., 2024; Collyer & Warren, 2009; Özkan & Mishra, 2019). This suggests that while Agile methods emphasize flexibility, they also require a strategic approach to planning that balances adaptability with sufficient foresight. Recent studies, such as those by Elkhatib et al. (2022) and Rahi et al. (2022), advocate for a hybrid approach that combines Agile and traditional planning methods to effectively manage project complexity and dynamism. This hybridization allows teams to leverage the strengths of both methodologies, enhancing their ability to respond to changes while maintaining a structured planning framework. Thus, the hypothesis proposed for investigation is that project planning significantly affects Agile project success.

Hypothesis 3: A key aspect of Agile methods is planning during execution

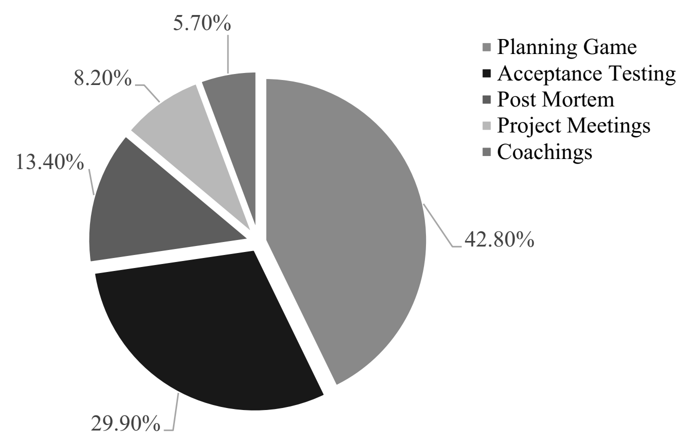

Exhibit 2. Customer Effort Distribution (%) for XP Projects, After Koskela and Abrahamsson (2004)

Project Characteristics

As we proceed, we will examine how specific agile project characteristics influence project success. It is crucial to recognize that each project is unique. While traditional project management literature often treats all projects as homogenous entities, each project possesses distinct characteristics that must be considered to develop an effective approach (Ciric Lalic et al., 2022). Project complexity, measured by various criteria, can be exacerbated by changes such as using emerging technologies or the introduction of new features (Aamer et al., 2024; Lappi et al., 2018).

Quality of goals and vision statement

The quality of goals and vision statements is critical in Agile project management, as they provide direction and facilitate informed decision-making. J. K. Pinto and Prescott (1990) demonstrated that a clear vision positively influences planning efforts and project success. However, the relationship between vision quality and Agile success is nuanced.

While Ćirić et al. (2022) argue that a strong vision energizes teams and fosters collaboration, they also acknowledge that aligning diverse stakeholder visions can be challenging in larger Agile projects. Moreover, Naslund and Kale (2020) emphasize that cultural factors and top management support are often more critical than the vision itself in determining the success of Agile transformations. This suggests that while a clear vision is beneficial, its effectiveness is contingent upon the organizational context and the ability of leadership to communicate and adapt that vision (Koch & Schermuly, 2021; Siddique & Hussein, 2022).

Furthermore, the role of leadership in reinforcing the vision is paramount, especially during periods of uncertainty. Koch and Schermuly (2021) highlight that effective leadership behaviors, such as articulating the mission and vision, are vital in Agile contexts where teams face external pressures. Additionally, Slåtten et al. (2021) found that organizational vision integration is positively related to employee performance, indicating that a well-communicated vision can enhance individual contributions to project success. However, the concept of "de-projectification" in Agile environments raises questions about the relevance of traditional vision statements, as the focus shifts toward adaptability and responsiveness rather than adherence to a fixed vision (Nesheim, 2023). Thus, while a clear vision is essential for Agile success, its impact is moderated by organizational culture, leadership effectiveness, and the dynamic nature of Agile practices.

Hypothesis 4: The relationship between Agile and success is stronger with a clear vision and goals.

Team characteristics

The characteristics of project teams are pivotal in determining the success of Agile projects (ForouzeshNejad et al., 2024). Agile methodologies emphasize the importance of self-organizing teams and effective communication, which are essential for fostering a motivating work environment built on trust and support (Dingsøyr et al., 2018). The Agile Manifesto underscores these principles, asserting that a motivated team is fundamental to delivering high-quality software (Dingsøyr et al., 2018).

Recent research supports this notion, indicating that Agile practices are particularly effective in smaller, co-located teams, where communication is more straightforward and collaboration is enhanced (Ciric Lalic et al., 2022; Elkhatib et al., 2022). However, the traditional view that Agile is only suitable for small teams has been challenged by more recent studies. For instance, Paasivaara et al. (2018) argue that Agile methodologies can facilitate communication and feedback even in larger, distributed teams, provided that appropriate frameworks and practices are in place.

Uludağ et al. (2021) further support this by demonstrating that Agile can be successfully implemented in large, distributed projects, although careful planning is necessary to mitigate potential communication challenges. This perspective is echoed by Jørgensen (2018), who notes that while Agile methods were initially designed for small teams, organizations have begun to adapt these practices for larger-scale projects, albeit with inherent challenges related to coordination and communication.

Team experience is another critical factor influencing Agile project success. Chow and Cao (2008) and Shenhar et al. (2001) highlighted the importance of team experience for Agile and general project success. While specific thresholds for team experience vary, it is generally suggested that having a significant proportion of experienced team members can lead to better project results (Simpson et al., 2024). Recent studies, such as those by Demirel et al. (2017) and Noll et al. (2017), further highlight the importance of team composition and the presence of high-caliber team members as critical success factors in Agile projects.

In summary, while smaller, co-located teams may offer advantages in Agile project success, recent literature indicates that Agile methodologies can be effectively scaled to larger teams and distributed environments, provided that communication strategies and team competencies are adequately addressed. Therefore, we propose the following hypotheses:

Hypothesis 5: The relationship between Agile and success is stronger with smaller teams than with larger teams.

Hypothesis 6: The relationship between Agile and success is stronger with experienced teams.

Customer involvement

Customer involvement is a cornerstone of Agile methodologies, as emphasized in the Agile Manifesto, which prioritizes "Customer collaboration over contract negotiation" (Dingsøyr et al., 2018). This customer-centric approach contrasts sharply with traditional plan-driven development models, which often limit customer interaction to the initial phases of a project (Park & Cho, 2022).

The significance of customer involvement in Agile practices is underscored by various researchers who highlight it as a critical factor for project success. For instance, Sankhe et al. (2022) noted that Agile methodologies, such as Scrum and

Extreme Programming (XP), inherently require continuous customer engagement throughout the development process, including planning, feedback, and review stages. This ongoing collaboration is essential for ensuring that the final product aligns with customer expectations and needs (Mann & Manurer, 2005; Rahy & Bass, 2022). The principles articulated by Dingsøyr et al. (2018) further reinforce this notion, asserting that satisfying the customer through early and continuous delivery of valuable software is paramount. In line with this, Simpson et al. (2024) emphasize that in XP practices, customer approval is necessary before each small release, highlighting the iterative nature of Agile development and the importance of customer feedback in shaping the product. Moreover, Almeida and Espinheira (2021) caution that without committed and knowledgeable customer participants, the likelihood of successful product transition diminishes significantly, even if the developed product meets initial satisfaction. This assertion is echoed by recent studies that demonstrate how customer involvement directly influences project outcomes, with high levels of engagement correlating with increased satisfaction and project success (Julià et al., 2021; Palfreyman & Morton, 2022).

The evidence strongly supports the hypothesis that high levels of customer involvement enhance the success of Agile projects. The iterative feedback loops facilitated by Agile methodologies not only ensure that the product meets customer needs but also foster a collaborative environment that can adapt to changing requirements. This adaptability is crucial in today's fast-paced development landscape, where customer expectations are continually evolving (Rahy & Bass, 2022).

Hypothesis 7: The relationship between Agile and success is stronger with high stakeholder involvement.

Organization dynamics – Co-location

Co-location is widely recognized as a significant factor influencing Agile project success. Frequent, face-to-face communication is often cited as crucial for achieving project success in Agile environments (Hofman et al., 2023; Junker et al., 2022). As Dingsøyr and Moe (2014) note, Agile methods were originally designed for small, co-located teams where easy access to users and business experts is paramount. This foundational principle suggests that while remote Agile practices can be effective, they may not fully capture the benefits of co-location.

Shared leadership and trust, particularly fostered in co-located settings, is reported enhancing productivity and project outcomes (Hofman et al., 2023; Junker et al., 2022). For instance, Junker et al. (2022), demonstrated that Agile work practices can positively influence team performance through proactive norms, which are more easily cultivated in co-located environments. However, research suggests that effective communication can occur in distributed teams with robust evaluation and feedback mechanisms. Werder and Maedche (2018) found that with the right systems, teams can maintain efficient communication across different locations, including international contexts. Lee et al. (2024) emphasizes the importance of perceived transparency in communication processes, suggesting that it can enhance project quality regardless of physical proximity. Furthermore, social Agile practices have been shown to facilitate both direct and indirect communication, mitigating the challenges of geographical dispersion (Hummel et al., 2015).

Despite these insights, our hypothesis remains that the relationship between Agile methodologies and project success is stronger in co-located teams. Face-to-face interactions foster a level of engagement and immediacy in communication that is difficult to replicate in remote settings. While Agile practices can be adapted to remote environments, the advantages of co-location, such as enhanced communication, trust, and shared leadership, continue to play a pivotal role in Agile project success. Therefore, we propose the following hypothesis:

Hypothesis 8: The relationship between Agile and success is stronger with co-located teams.

Methodology

Our research takes a post-positivist stance in recognizing the correlation between project planning and project success. Post-positivism is an epistemological approach that occupies a middle ground between positivism, which aims to find completely objective solutions to research questions, and phenomenology, which acknowledges the subjective nature of all experiences (Trochim & Donnelly, 2001; William, 2024). As perception and observation can be influenced by subjective opinion, research results cannot be entirely objective. Additionally, certain concepts, such as project success, may not be entirely quantifiable and may be impacted by the subjective judgment of project participants and sponsors. Thus, post-positivism recognizes that while positivism may not provide a complete picture in business research, its insights can still be valuable.

This study gathered data on quantities such as success measures, the use of agile and project characteristics that moderate success, see Exhibit 3. A quantitative approach was taken to analyze this data. From previous analysis, there was a clear relationship between Agile use and success. The simple approach taken here was to do subgroup analysis of the relationship for the various states of the moderating variables. The regression relationship will hold unless the moderating variable interferes with the relationship between Agile and success as shown by a regression analysis. In those cases, the regression will no longer show as statistically significant.

Survey

Given that this study, like many others investigating project success, employed questionnaires and interviews, the findings are subject to the participants' subjective perceptions of the projects. Consequently, the current paper primarily focuses on the perceived project success as reported by the respondents. To assess this factor, the survey items were derived from a combination of success dimensions outlined by previous studies as outlined in Appendix.

Data were gathered from project management practitioners affiliated with either the Project Management Institute (PMI) or various LinkedIn project management groups. Given the overlapping membership between these

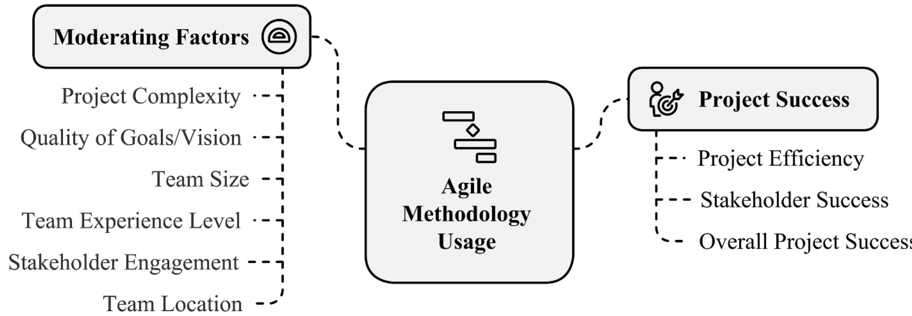

Exhibit 3. Main Research Constructs

groups and the inherent uncertainty associated with open surveys in gauging visibility, the precise extent of the population remains indeterminate. A total of 865 people started the survey with 859 providing information on at least one successful project. The largest group of respondents were project managers (52%) though respondents also included team members (7%) and senior management (6%). There were 17% of respondents who did not specify their role. Each participant was asked to provide data on two projects, one more successful and another less successful. However, not all participants entered data for two projects; therefore, the total number of projects was 1539. After removal of outliers and bad data, the usable total was 1,386 projects. A thorough examination of the distribution of success ranges revealed adherence to normality criteria, thereby ensuring the statistical robustness of the dataset.

A large percentage of respondents were from the USA (36%), but there is still a global level of participation in the survey with several countries having 10 or more respondents: India (6.9%), Canada (6.7 %), Australia (2.2 %), Spain (1.7%), Brazil (1.3%), Singapore (1.3%), and Germany (1%). 21.3% chose not to answer the question. Overall, there was a good global representation with respondents from more than sixty countries.

Participant Error Issues

Participant errors can affect the accuracy of survey results, as highlighted by previous studies (Celhay et al., 2022; Cooper et al., 2006). In this study, some participants may have lacked access to project data, resulting in unrealistic responses, and outliers were removed during analysis. Where there were problems in survey responses to the agile effort questions, the data was repaired based on responses to other questions indicating the use of Agile within the project. The resulting data consistency was confirmed. An Agile factor was calculated using the repaired data and respondents' assessment of Agile use.

Success Measures

There were three questions that measured project efficiency, and four that measured overall project success. A confirmatory factors analysis using normalized varimax rotation was completed for these success measures followed by a Cronbach alpha analysis. All measures of success showed a high Cronbach alpha score which shows they are all correlated. This is in agreement with the review of the success literature and the view that all success measures are to some extent correlated (Butler et al., 2020; Ciric Lalic et al., 2022; Jørgensen, 2018).

Based on these findings, there is insufficient justification to exclude any of the items from the analysis. The average inter-item correlation coefficient stands at 0.905, and the marginal improvement in alpha by removing the "project budget goals" item suggests its retention. For analytical purposes, the success measures were aggregated into three distinct success factors:

- Efficiency Factor: Comprising the mean ratings for the project's performance in meeting budget, time, and scope objectives.
- Stakeholder Success Factor: Encompassing the mean ratings for the project's success as perceived by project sponsors, stakeholders, project team members, clients, and end users.
- Success Factor: Consisting of the mean ratings from the preceding seven responses, in addition to an overall ranking of the project's success.

The bulk of the analysis centered on these three factors along with the overarching question pertaining to the project's overall success. Both confirmatory factor analysis and Cronbach's Alpha assessments were conducted to further scrutinize the data. Most of the analysis was based upon these three factors and the question asking respondents to rate the project's overall success.

Other Measures

To facilitate the analysis of the planning effort during execution as a percentage of the total, the following indexes were calculated:

$$Planning\ Effort\ Index = \frac{Total\ efforts\ expended\ during\ theplanning\ phase}{The\ total\ project\ effort\ (inpersondays)} \quad (1)$$

$$Agile\ Planning\ Effort\ Index = \frac{Total\ efforts\ expended\ onplanning\ during\ the\ execution\ phase}{The\ total\ project\ effort\ (inpersondays)} \quad (2)$$

These indexes were used to identify outliers. Projects that spent more than 60% of their time planning, or were canceled or incomplete, were excluded from the analysis. Projects with valid agile effort indexes and planning characteristics totaled 1002. Analysis was completed with success factors as the dependent variable and agile factor as the independent factor:

- Agile Factor: Combination of respondents' assessment of Agile use in the project along with the Agile Planning Effort Index.

Results and Discussion

The Impact of Agile on Success

According to Exhibit 4, approximately 935 projects (67.5% of the original 1386 projects) reported incorporating some form of agile or iterative component.

To further validate the impact of Agile methodologies, a one-way analysis of variance (ANOVA) was conducted, examining the relationship between the percentage of Agile/iterative usage and project success measures. The ANOVA analysis provided additional evidence supporting the positive influence of Agile on project success, see Exhibit 5.

Exhibit 5 provides valuable insights into the relationship between Agile methodology usage, planning effort, and project success factors. As hypothesized, a higher percentage of Agile/iterative methodology usage is associated with higher ratings for stakeholder success and overall project success. This finding aligns with previous research highlighting the positive impact of Agile practices on customer satisfaction and project outcomes (Al Jabri et al., 2023; Gerona & Ocampo, 2023). The results also highlight effective planning and customer involvement could play significant roles in achieving favorable project outcomes that is also inline with previous research.

In terms of stakeholder success factor, the p(F) value shows statistical significance between agile use and the three success measures with the exception of Efficiency which is somewhat over the threshold. $p < .05$ is acceptable and this table shows values substantially lower. We can therefore reject the null hypothesis. The amount of planning during execution is also significantly related to methodology assessment by respondents (% use of Agile) as expected. This finding aligns with previous research highlighting the positive impact of Agile practices on customer satisfaction and project outcomes (Gerona & Ocampo, 2023; Koskela & Abrahamsson, 2004)

Serrador and Pinto (2015) found that there is no difference in the mean upfront planning amounts between agile and traditional projects. However, if substantial planning is done during execution, then agile projects tend to do more planning overall. To address the issue of using a single item measure, an index was created by combining a question on the project's agilities with a measure of how much planning was done in the execution phase. A factor was defined using this index.

Combined agile factor = mean of the following two responses as a summated scale of normalized values:

(1) Methodology type
(2) Agile planning index

Where Agile planning index is defined as follows:

$$Agile\ Planning\ Index = \frac{Total\ efforts\ expended\ on\ planning\ after\ the\ planning\ phase}{The\ total\ project\ effort\ (inpersondays)} \tag{3}$$

Both items are taken as a measure of the "agileness" of the project. The first is based on respondent assessment of how much agile process is used in the project; the second on how much planning is completed during execution which is a feature of agile projects (Al Jabri et al., 2023; Elkhatib et al., 2022). As found by Serrador and Turner (2015), the relationship between planning effort and success is an inverted U and too much planning can also have a detrimental impact on project success, and this is a caveat for the linear analysis that follows.

The analysis in Exhibit 6 demonstrates a positive correlation between Agile methodology usage and project success, supporting the hypothesis that Agile projects can outperform traditional projects. To further explore this relationship and identify potential moderating factors, a subgroup analysis was conducted. This analysis aims to examine how the effectiveness of Agile methods varies across different project contexts, such as project goal and vision, complexity, team size, location and experience, use of technology, and stakeholder engagement. By understanding these nuances, organizations can make informed decisions about the suitability of Agile methodologies for specific projects.

The findings regarding the impact of goal and vision statement quality on the relationship between Agile use and project success provide some interesting insights, See Exhibit 7.

Exhibit 4. Frequency Table for Methodology Type

Percentage Agile/Iterative	Count	Cumulative – Count	Percentage
80–100%	80	80	5.7
60–79%	152	232	10.9
40–59%	347	579	25.0
20–39%	162	741	11.6
1–19%	194	935	13.9
0%	451	1386	32.5

Exhibit 5. Methodology Type Means with ANOVA Analysis

	Means and ANOVA					
Percentage Agile/Iterative	Upfront planning effort index	Agile planning effort index	Success Factor	Stakeholder Success Factor	Efficiency factor	Valid N
80–100%	0.161	0.149	3.569	3.638	4.821	80
60–79%	0.147	0.138	3.475	3.567	4.664	152
40–59%	0.164	0.132	3.500	3.544	4.793	347
20–39%	0.135	0.101	3.361	3.414	4.638	162
1–19%	0.150	0.091	3.173	3.179	4.460	194
0%	0.154	0.048	3.223	3.208	4.582	451
All Groups	0.153	0.105	3.384	3.376	4.647	1386
p(F)	0.173	0.000	0.000	0.000	0.087	

Exhibit 6. Standard Linear Regression Analysis of Combined Agile Factor vs. Project Success Rating

Regression for project success rating
$R = .138$ $R^2 = .019$ Adjusted $R^2 = .018$ $N = 1002$

	Beta	B	*p*-level
Intercept			0.0000
Combined Agile measure	0.138	0.031	0.00001

Exhibit 7. The Impact of Agile Factor on Success by Quality of Goals and Vision Statement

Quality of goals/vision statement	Number of Projects	Beta	B	R^2	*p*-value
Excellent	189	0.021	0.011	0.000	0.772
Good	567	0.116	0.072	**0.014**	**0.005**
Poor	196	0.181	0.106	**0.033**	**0.010**
Very Poor/Not used	38	0.291	0.186	0.085	0.055

The highest impact was for projects that had neither very good nor very poor goals/vision statements. However, very poor was also close to significance with a high R^2. This finding appears to contradict the assertions made by Ciric Lalic et al. (2022) who argued that a clear vision is crucial for project success. However, it aligns with the observations of Naslund and Kale (2020), who emphasized that cultural factors and leadership support may be more important than the vision itself in determining the success of Agile transformations.

One possible explanation for these results is that in projects with extremely well-defined goals and vision, the benefits of Agile's flexibility and adaptability may be less pronounced, as the project can already be effectively managed using more traditional, plan-driven approaches. Conversely, in projects with very poor or absent goal and vision statements, the challenges of aligning diverse stakeholder perspectives may limit the ability of Agile practices to enhance project success (Ciric Lalic et al., 2022). This suggests that the effectiveness of Agile is not solely dependent on the clarity of the initial vision, but rather on the ability of the methodology to adapt to changing circumstances, even in the absence of a well-articulated project roadmap.

Project complexity was considered as the next part of analysis, and results are shown in Exhibit 8. As demonstrated in the table, high and medium complexity projects benefit significantly from Agile approaches, with R-squared values of 0.035 and 0.016 respectively. This suggests that the flexibility and iterative nature of Agile allow teams to effectively navigate the challenges posed by complex projects, leading to improved success rates, that is in line with previous studies (Barlow et al., 2011; Serrador & Pinto, 2015). For low complex projects, it can be argued that traditional, plan-driven methodologies may be equally or more effective than Agile approaches. This is because less complex projects may not necessitate the same level of structure and continuous adaptation inherent in Agile practices (Demirel et al., 2017; Tominc et al., 2023).

The project size was also examined as part of the analysis, and, as demonstrated in Exhibit 9, the results showed Agile works better in smaller teams. Teams larger than 30 do not show clear benefits from Agile methods while those that are smaller clearly do. This is likely due to the inherent emphasis Agile places on self-organizing, cross-functional teams, which can be more challenging to coordinate and manage in larger project settings. The challenges of scaling Agile to larger teams, as discussed by Jørgensen (2018) and Uludağ et al. (2021), are reflected in the finding that teams larger than 30 do not exhibit clear benefits from Agile approaches.

Exhibit 8. The Impact of Agile Factor on Success by the Complexity of the Project

Complexity of the project	Number of Projects	Beta	B	R^2	*p*-value
Low	71	0.129	0.084	0.016	0.278
Medium	510	0.128	0.078	**0.016**	**0.004**
High	415	0.187	0.132	**0.035**	**0.000**

Exhibit 9. The Impact of Agile Factor on Success by Project Team Size

Project team size	Number of Projects	Beta	B	R^2	*p*-value
15	222	0.189	0.120	**0.036**	**0.005**
6–15	373	0.139	0.093	**0.019**	**0.007**
16–30	184	0.205	0.136	**0.042**	**0.005**
31–50	89	0.119	0.074	0.014	0.261
51–100	76	0.020	0.013	0.000	0.864
101–500	39	0.123	0.091	0.015	0.444
500+	5	0.228	0.221	0.052	0.622

Teams experience was also investigated as shown in Exhibit 10. Agile appears to have the greatest impact on medium experience teams. Highly experienced and low experience teams do not seem to gain as much from Agile methodologies, though there does appear to be some gain. This is most likely because highly experienced teams can deliver successful projects without advanced methodologies such as Agile. On the other hand, inexperienced teams may benefit from a more structured approach provided by a waterfall type methodology (Demirel et al., 2017; Noll et al., 2017).

For degree of stakeholder engagement (Exhibit 11), projects with very engaged stakeholders do not appear to gain much from Agile. However, all other projects seem to have a clear benefit, though "not at all engaged" projects are not significant, but there is a high R^2. This contradicts the emphasis placed on customer collaboration in the Agile Manifesto and prior research, such as the work by Koskela and Abrahamsson (2004) and Sankhe et al. (2022), which highlighted the importance of continuous customer involvement in Agile projects. This is likely due to the very low number of projects, but speculation suggests that projects with poorly engaged stakeholders also benefit from agile. Overall, the analysis of stakeholder engagement provides a nuanced perspective,

Exhibit 10. The Impact of Agile Factor on Success by Experience Level of Team

Experience level of team	Number of Projects	Beta	B	R^2	*p*-value
High	350	0.101	0.060	0.010	0.057
Medium	539	0.174	0.114	**0.030**	**0.000**
Low	116	0.152	0.094	0.023	0.100

Exhibit 11. The Impact of Agile Factor on Success by Degree of Stakeholder Engagement

Degree of stakeholder engagement	Number of Projects	Beta	B	R^2	*p*-value
Very Engaged	398	0.040	0.024	0.001	0.430
Somewhat Engaged	450	0.144	0.088	**0.021**	**0.002**
Mostly not Engaged	130	0.292	0.179	**0.085**	**0.001**
Not at all Engaged	16	0.372	0.230	0.139	0.128

highlighting that Agile approaches may be beneficial across a range of stakeholder involvement levels, rather than being solely dependent on high stakeholder engagement as previous literature has suggested. This adds an important qualification to the understanding of how Agile methodologies interact with project context and characteristics

The analysis of hypothesis H8, as presented in Exhibit 12, reveals distinct responses to Agile methodologies based on teams' geographical distribution. Teams with an international composition, spanning multiple countries, or a national presence across multiple cities, notably demonstrate a clear benefit from Agile practices. Conversely, teams that are collocated within a single city or region do not exhibit a pronounced advantage from Agile methodologies. This observation suggests that the heightened communication facilitated by physical proximity in collocated teams may mitigate the perceived benefits of Agile methodologies. This observation contradicts the assertions made by Dybå and Dingsøyr (2008) and the principles outlined in the Agile Manifesto, which emphasize the importance of face-to-face communication and co-location for Agile success. However, the current findings align with more recent research, which has challenged the notion that Agile is only suitable for small, co-located teams. For instance, Paasivaara et al. (2018) argue that Agile methodologies can facilitate communication and feedback even in larger, distributed teams, provided that appropriate frameworks and practices are in place. Similarly, Uludağ et al. (2021) demonstrate that Agile can be successfully implemented in large, distributed projects, although careful planning is necessary to mitigate potential communication challenges. In light of these findings, hypothesis H8 is deemed false. The results indicate that teams need not be collocated to derive benefits from Agile methodologies. Instead, the efficacy of Agile appears to be more pronounced in geographically dispersed teams, whether internationally or nationally distributed.

As shown in Exhibit 13 and 14, the study found strong support for the overall effectiveness of Agile methodologies, with Hypothesis 1 being confirmed - Agile projects were found to be more successful than traditional projects. However, the relationship between Agile and project success was moderated by several contextual factors. Contrary to Hypothesis 2, the benefits of Agile were not limited to high complexity projects but were observed across projects of varying complexity levels. Consistent with Hypothesis 3, the study highlighted the importance of planning during Agile execution as a key driver of success. Interestingly, Hypothesis 4 was not supported, indicating that Agile can be beneficial even when project goals and vision are not clearly defined. Team size emerged as an important factor, with Hypothesis 5 being confirmed-Agile Agile was found to be more effective for smaller project teams compared to larger ones.

Exhibit 12. The Impact of Agile Factor on Success by Location of Team

Location of Team Members	Number of Projects	Beta	B	R^2	*p*-value
International	350	0.158	0.108	0.309	**0.040**
National	539	0.205	0.136	0.257	**0.002**
One city or region	116	0.126	0.080	0.418	0.134

In contrast, Hypothesis 6 was not supported, as medium-experienced teams seemed to benefit the most from Agile, rather than highly experienced teams. The study also challenged conventional wisdom regarding the role of stakeholder engagement. Contrary to Hypothesis 7, low stakeholder involvement did not preclude teams from deriving benefits from Agile. Similarly, Hypothesis 8 was not supported, as non-collocated teams were also found to benefit from Agile practices.

Overall, the findings suggest that while Agile methodologies offer significant advantages over traditional approaches, their effectiveness is contingent on carefully considering the unique characteristics of each project context. Organizations should adopt a nuanced approach when implementing Agile, tailoring its application to the specific needs and constraints of their projects

Theoretical Implications

The subgroup analysis provides valuable insights into the contextual factors that influence the efficacy of Agile methodologies. These findings align with and expand upon the theoretical background presented earlier. Consistent with the literature, the analysis indicates that Agile is particularly effective in projects characterized by moderate team experience, moderate-to-low stakeholder engagement, medium-to-high complexity, and smaller team sizes (Chow & Cao, 2008; Demirel et al., 2017; Junker et al., 2022; Shenhar et al., 2001). This supports the notion that Agile practices, such as iterative development, continuous feedback, and self-organizing teams, are well-suited for navigating the challenges of complex, uncertain environments (Koch & Schermuly, 2021; Muhammad et al., 2021).

Notably, the success of Agile in projects with initially unclear goals contradicts the emphasis on clear vision and goals posited by Lalić et al. (2022). This suggests that Agile's adaptability and flexibility can compensate for ambiguous project requirements, aligning with the perspective that Agile methodologies excel at responding to change (Dingsøyr et al., 2018; Rahy & Bass, 2022). However, the findings also caution against assuming the universal benefits of Agile. The lack of Agile's impact in low-complexity projects, those with clearly defined goals, highly engaged stakeholders, large teams, and inexperienced teams contradicts the notion of Agile's suitability for all project contexts (Barlow et al., 2011; Dybå & Dingsøyr, 2008; Serrador & Pinto, 2015). This nuance underscores the importance of carefully evaluating project characteristics when selecting an appropriate management approach, as advocated by Demirel et al. (2017) and Shenhar et al. (2001)

The theoretical implications of these findings further elucidate the factors contributing to Agile's growing popularity. The emphasis on continuous communication with stakeholders and ongoing development aligns with the principles of the Agile Manifesto, which prioritize customer collaboration and responding to change (Beck et al., 2001; Dingsøyr et al., 2018). Additionally, the benefits observed for small and moderately experienced teams corroborate the original design of Agile methods for such environments (Dybå & Dingsøyr, 2008; Jørgensen, 2018). Conversely, the limitations

Exhibit 13. Summary of The Results of Hypotheses Tests

Hypothesis	Description of Hypothesis	Results
H1	Agile projects are more successful than traditional projects	Confirmed
H2	High complexity projects are best suited to an Agile environment.	Not Confirmed – all project benefit
H3	A key aspect of agile methods is planning during execution.	Confirmed
H4	The relationship between Agile and success is stronger with a clear vision and goals.	Not Confirmed – Agile can help with less clear vision/goals
H5	The relationship between Agile and success is stronger with smaller teams than with larger teams.	Confirmed
H6	The relationship between Agile and success is stronger with experienced teams.	Not Confirmed – medium experience teams benefit most
H7	The relationship between Agile and success is stronger with Projects require high stakeholder involvement to benefit from agile.	Not confirmed – low stakeholder engagement teams benefit
H8	The relationship between Agile and success is stronger with Agile processes requiring co-located teams to achieve benefits.	Not confirmed – non co-located teams also benefit from Agile

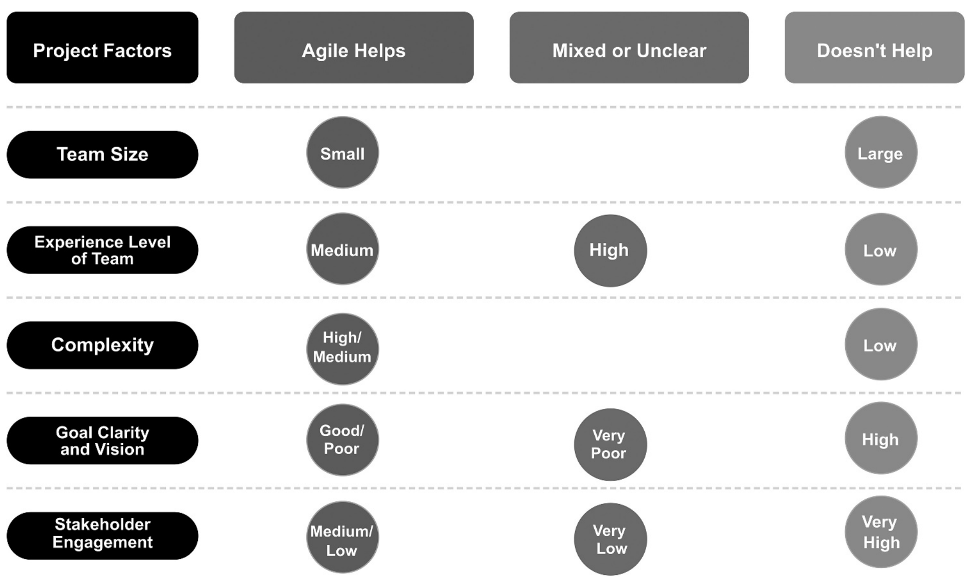

Exhibit 14. Impact of Agile with Different Project Characteristics

of Agile in highly experienced, large, or low-complexity teams underscore the need for contextual considerations when adopting Agile, as suggested by Nuottila et al. (2022). These findings contribute to the ongoing evolution of project management theory, reinforcing the need for a contingency-based perspective that accounts for the unique demands of each project environment.

Practical Implications

The findings of this study offer several practical implications for organizations considering Agile methodologies. Firstly, the research highlights that Agile can be effective in areas not previously considered a strong fit, such as distributed teams, less experienced teams, and projects with varying complexity and stakeholder engagement (Dybå & Dingsøyr, 2008; Paulk, 2002). Organizations should explore Agile's potential across a broader range of project contexts.

Furthermore, the research indicates Agile can be particularly beneficial for high-risk projects with low stakeholder involvement, unclear vision, or distributed teams, which often struggle with traditional methodologies (Barlow et al., 2011; Gerona & Ocampo, 2023).

Additionally, for projects with large teams or highly experienced members, project managers should focus on enhancing communication, coordination, and knowledge sharing to address the limitations identified in this study. Strategies such as implementing robust collaboration tools, fostering a shared understanding of Agile principles, and providing targeted training can help large or experienced teams derive greater benefits from Agile methodologies. For projects with unclear goals or vision, organizations should consider using Agile methodologies to help foster adaptability and responsiveness, rather than relying solely on upfront planning. Agile's iterative nature and emphasis on continuous feedback can help teams align the project with evolving business needs, even in the absence of a clearly defined initial vision. In projects with low stakeholder engagement, Agile practices such as regular demos and retrospectives can help reengage stakeholders and ensure the project remains aligned with their needs. Project managers should proactively seek stakeholder involvement throughout the project lifecycle, rather than assuming high engagement is a prerequisite for Agile success. When working with smaller, less experienced teams, project managers should focus on building the necessary skills and capabilities to effectively implement Agile practices. This may include providing Agile training, mentoring, and opportunities for the team to gain practical experience with the methodology. Managers could also consider using a Hybrid Agile approach to add more structure for some teams. Hybrid approaches have been shown to be as effect as Agile approaches when designed correctly (Gemino et al., 2021).

However, the study cautions against assuming universal Agile benefits, especially in low-complexity projects, those with clearly defined goals, large teams, or inexperienced members. In these scenarios, organizations should carefully evaluate Agile's appropriateness and consider hybrid approaches (Demirel et al., 2017; Jørgensen, 2018).

Overall, the practical implications encourage a more nuanced, contextual approach to Agile implementation, rather than viewing it as a one-size-fits-all solution. Understanding the specific project characteristics that enhance or limit Agile's effectiveness can help decision-makers make informed choices for optimal project success.

Conclusion

In conclusion, this study underscores the effectiveness of Agile methodologies while highlighting the importance of context in determining their success. Agile's adaptability proves advantageous in projects with moderate team experience, smaller teams, medium-to-high complexity, low stakeholder engagement, or unclear initial goals, but its benefits are less pronounced in low-complexity projects, large teams, or scenarios with high stakeholder involvement. These findings challenge assumptions about Agile's universal applicability or unsuitability, emphasizing the benefit of organizations carefully evaluating project-specific factors before implementation.

Limitations and Future Research

Ensuring adequate survey participation has become increasingly challenging in recent years, as noted by Fellows and Liu (2021), who highlight the growing difficulty in data collection due to the rising number of research projects. As with survey-based research in general, we rely on the information and judgment of the participants regarding the success of the project, project characteristics, and the use of Agile. We depend on the respondents having a good grasp of what percentage of project processes were Agile processes vs. traditional processes as an example. The respondents were typically project managers or senior team members, so we were assuming intimate knowledge of how the project was run. Percentage (%) of Agile use refers to methods. For example, a project may use some Agile methods like iterations and standup meeting and some waterfall methods such as WBS and use cases. We depend on the respondents to accurately judge the percentages.

Some of the R^2 relationships reported in our study were found to be relatively small, which is not uncommon in project management research. However, it's important to recognize that certain R^2 values exhibited significant strength even before conducting further analyses. The robustness of the reported p-values and the substantial size of our datasets lend support to the validity of these relationships. Control variables were not used in the regression analyses as the main focus was on finding effects rather than their strength. This limitation could be addressed in future research through the implementation of Multiple Hypothesis Regression Analysis (MHRA). That could also uncover stronger R^2 values. Future research could further explore these relationships in more detail as well as the interaction between other project characteristics and the use of Agile.

Disclosure Statement

No potential conflict of interest was reported by the author(s).

Notes on contributors

Pedro M. Serrador, PMP, ACP, P.Eng., MBA, PhD is a writer and researcher on Agile, AI and project success. As well as having more than 20 years of industry experience, he is an adjunct professor at the Northeastern University and University of Toronto. He an author of books and articles on project management and was the recipient various awards for his academic writing including the 2022 most cited paper award from Project Management Journal.

Adel A. Zadeh is a Professor of Project/Construction Management at Northeastern University in Toronto. He holds a Ph.D. in Engineering from Cambridge University, UK, and has over a decade of combined industrial and academic experience. He is also serving as the Director of the College of Professional Studies in Toronto. Dr. Zadeh is actively engaged in various research projects, including organizational transformation, AI in project and construction management, agile and hybrid project management, sustainable construction, lean construction, and green buildings.

ORCID

Adel A. Zadeh PhD http://orcid.org/0000-0001-7826-1460

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Appendix. Survey questions used in this study

ID	Question	Response ranges	Reference
1	*Project team size*: How large was the project team (full time staff equivalent)	*7 point scale*: • 1–5 • 6–15 • 16–30 • 31–50 • 51–100 • 101–500 • 501+	Crawford et al. (2004); Uludağ et al. (2021)
2	*Project complexity*: Rate the complexity of the project	*3 point scale*: • Low • Medium • High	Dulewicz et al. (2005); Tominc et al. (2023)
3	*Project length*: What was the project length (full lifecycle)	*3 point scale*: • <1 year • 1 to 3 years • >3 years	Kieffer et al. (2017)
4	*WBS*: Rate the detail level of the WBS (work breakdown structure) used the project	*4 point scale*: • Excellent • Good • Poor • Very Poor/Not used	Baumgartner et al. (2021)
5	*Vision Statement*: Rate the applicability/quality of the vision statement or project goal definition for the project.	*4 point scale*: • Excellent • Good • Poor • Very Poor/Not used	Ciric Lalic et al. (2022)
6	*Novelty to the organization*: How new is this type of project to the organization?	*4 point scale*: • Very • Somewhat • Mostly not • Not at all	Ghosh and Wu (2023)
7	*Internal vs. vendor based*: What percentage of project was completed by vendors?	*6 point scale*: • 80–100% • 60–79% • 40–59% • 20–39% • 1–19% • 0%	Crawford et al. (2004); Rajagopalan and Mathew (2016)
8	*Industry*: What industry was the project in? (choose the best fit)	• Education • Retail • High technology • Financial services • Manufacturing • Utilities • Health care • Government • Professional services • Telecommunications • Construction • Other	Shenhar et al. (2001), Premkumar and King (1991), Catersels et al. (2010)
9	*Geographic location of project*: What was the geographic location of project?	• Europe • Indian subcontinent • Russia and FSU • Far East • North America • Australasia • Latin America • Pacific • Middle East • Arctic and Antarctic • Africa sub-Sahara	Hofman et al. (2023); Junker et al. (2022)
10	*Local vs. remote team*: Where were the team members located? Choose the option that best fits the majority of team members.	*3 point scale*: • One city or region • National • International	Hofman et al. (2023); Junker et al. (2022)

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ID	Question	Response ranges	Reference
11	*Level of use of technology*: Low tech indicates none or very mature technology where super high-tech indicates the use or development of completely new technology.	*4 point scale*: • Low-tech • Medium-tech • High-tech • Super high-tech	Shenhar et al. (2001); Zakrzewska et al. (2024)
12	*New product vs. maintenance*: Does this project involve developing a new product, installation or system or is it related to maintenance of what already exists?	*3 point scale*: • New product development • Product modifications • Maintenance only	Shenhar et al. (2001); Zakrzewska et al. (2024)
13	*Experience level of team*: How experienced was the project team?	*3 point scale*: • High, • Medium, • Low	Demirel et al. (2017); Simpson et al. (2024)
14	*Degree of stakeholder engagement*: How engaged were the key stakeholders for the project?	*4 point scale*: • Very • Somewhat • Mostly not • Not at all	Rahy and Bass (2022)
15	*Methodology type*: How much of the project was done using agile or iterative techniques? (100 = fully agile, 0 = fully waterfall, 50 = an equal mix of agile and waterfall techniques)	*6 point scale*: • 80–100% • 60–79% • 40–59% • 20–39% • 1–19% • 0%	Lappi et al. (2018)
16	*Project effort*: What was the total project effort (in person days)	Numerical – person days	Usman et al. (2018)
17	*Planning Phase effort*: What was the total effort expended on the planning phase (person days)? Planning is defined as everything before execution.	Numerical – person days	Usman et al. (2018)
18	*Total effort spent on planning over life of project*: For Agile/Iterative projects, what is the total effort expended in planning over the course of the project? This should include all planning down to weekly planning sessions.	Numerical – person days	Usman et al. (2018)
19	*Project Budge*: What was the overall project budget? (USD - please do the approximate conversion to US Dollars)	Numerical:Approximate in US dollars	Crawford et al. (2004); Lang et al. (2013)
20	*Project success rating– overall*: How do you rate the overall success of the project?	*5 point scale*: • failure, • not fully successful, • mixed, • successful, • very successful	Binboga and Altin Gumussoy (2024); RamadanDarwish and Rizk (2015)
21	*Project success rating– sponsor feedback*: How did the project sponsors and stakeholders rate the success of the project?	*5 point scale*: • failure, • not fully successful, • mixed, • successful, • very successful	RamadanDarwish and Rizk (2015); Binboga and Altin Gumussoy (2024)
22	*Success: Meeting budget goals*: How successful was the project in meeting project budget goals?	*7 point scale*: • >60% over budget, • 45–59% over budget, • 30–44% over budget, • 15–29% over budget, • 1–14% over budget, • on budget • under budget	RamadanDarwish and Rizk (2015); Binboga and Altin Gumussoy (2024)
23	*Success: Meeting timeline goals*: How successful was the project in meeting project time goals?	*7 point scale*: • >60% over time, • 45–59% over time, • 30–44% over time, • 15–29% over time, • 1–14% over time, • on time • ahead of schedule	RamadanDarwish and Rizk (2015); Binboga and Altin Gumussoy (2024)

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(Continued).

ID	Question	Response ranges	Reference
24	*Success: Meeting scope/requirements goals*: How successful was the project in meeting scope and requirements goals?	*7 point scale*: • >60% requirements missed, • 45–59% requirements missed, • 30–44% requirements missed, • 15–29% requirements missed, • 1–14% requirements missed, • requirements met • requirements exceeded	RamadanDarwish and Rizk (2015); Binboga and Altin Gumussoy (2024)
25	*How do you rate the project team's satisfaction with the project?*	*5 point scale*: • failure • not fully successful • mixed • successful • very successful	Huijgens et al. (2017); Müller and Turner (2007)
26	*How do you rate the client's satisfaction with the project's results?*	*5 point scale*: • failure • not fully successful • mixed • successful • very successful	Huijgens et al. (2017); Müller and Turner (2007)
27	*How do you rate the end users' satisfaction with the project's results?*	*5 point scale*: • failure • not fully successful • mixed • successful • very successful	Huijgens et al. (2017); Müller and Turner (2007)
Demographic data was collected for the respondents to the survey. The following information was requested:			
28	*Job function*: What is your job function?	*8 point scale*: • Project Team Member • Project Coordinator • Project Manager • Senior Project Manager • Program Manager • Senior Program/ Portfolio • Senior Manager • C Level Management	Crawford et al. (2004); Müller and Turner (2007)
29	*Age*: What is your age?	*5 point scale*: • <30 • 31–40 • 41–50 • 51–60 • 61+	Müller and Turner (2007)
30	*Years of experience in project management*: How many years of experience do you have in project management?	Numerical	
31	*Gender*: What is your gender?	M/F	Müller and Turner (2007)
32	*Project management certification*: What project management certifications do you possess? Check all that apply.	• PMP • Prince2 • IPMA • APMC • AAPM • AIPM • PgMP	Müller and Turner (2007)
33	What is your highest education level?	*4 point scale*: • High school • Bachelor's degree • Master's degree • Doctorate	Müller and Turner (2007)